



The Landscape Resilience Index (LRI)

A handbook to facilitate integrated landscape management

Ala Al Dwairi and Maria Ana Borges





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About IUCN

IUCN is a membership Union uniquely composed of both government and civil society organisations. It provides public, private and non-governmental organisations with the knowledge and tools that enable human progress, economic development and nature conservation to take place together.

Created in 1948, IUCN is now the world's largest and most diverse environmental network, harnessing the knowledge, resources and reach of more than 1,500 Member organisations and around 19,000 experts. It is a leading provider of conservation data, assessments and analysis. Its broad membership enables IUCN to fill the role of incubator and trusted repository of best practices, tools and international standards.

IUCN provides a neutral space in which diverse stakeholders including governments, NGOs, scientists, businesses, local communities, indigenous peoples organisations and others can work together to forge and implement solutions to environmental challenges and achieve sustainable development.

Working with many partners and supporters, IUCN implements a large and diverse portfolio of conservation projects worldwide. Combining the latest science with the traditional knowledge of local communities, these projects work to reverse habitat loss, restore ecosystems and improve people's well-being.

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About SUSTAIN

The Sustainability and Inclusion Strategy for Growth Corridors in Africa (SUSTAIN-Africa) is a flagship initiative of IUCN's Water and Wetlands Team. The first phase of SUSTAIN-Africa (2014–2020) focused on catalysing climate-resilient development that balances economic growth, social prosperity, and ecosystem stewardship in the Southern Agricultural Growth Corridor of Tanzania (SAGCOT) and the Zambezi Valley in Mozambique. This first phase established multi-stakeholder partnerships with businesses, governments, communities, and farmer groups through an integrated landscape management approach.

Building on these achievements, the second phase of SUSTAIN-Africa (2022–2026) included two complementary programmes:

- SUSTAIN Pro, which focused on the development of sustainable food systems and healthy productive landscapes in Tanzania and Mozambique
- SUSTAIN Eco, which promoted healthy ecosystems and prosperous communities in Tanzania.

www.sustaininitiative.org

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IUCN is particularly grateful to all the SUSTAIN-Africa stakeholders who contributed their time, knowledge

and experience throughout the process, including local communities, farmers, Water User Association members, government representatives, technical experts and other landscape actors. Their input was essential in ensuring that the LRI remained both technically sound and practical for use in real-world landscape contexts.

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Finally, a word of appreciation goes to Maria Ana Borges and Ala Al Dwairi for coordinating this engagement and developing the relevant documents, including this handbook.

About this handbook

This handbook is a practical manual for stakeholders in public institutions, private companies and development organisations seeking to generate usable intelligence for landscape-level investment and policy. By following the implementation steps laid out in this handbook, practitioners can bridge the gap between local action on the ground with broader national and international goals, such as the Nationally Determined Contributions (NDCs) and the Sustainable Development Goals (SDGs).

The primary audience of the handbook are landscape practitioners engaged in designing and managing integrated landscape management programmes. Its intent is to provide them with a tried and tested set of indicators, which together can measure progress

towards landscape resilience and support adaptive management for improved landscape outcomes. Secondary audiences to this guide are public and private decision-makers engaged in landscape partnerships or involved in implementing nature-based solutions at different levels.

The handbook presents the structure of the LRI, details the recommended sequence of implementation steps (from scoping to baseline profiling), and provides decision-making pathways for applying the results in practice. Finally, it includes two baseline assessments from the SUSTAIN-Africa initiative in Tanzania and Mozambique to illustrate how the LRI is applied to a landscape context.

Acronyms

ADEM	Manica Agency for Economic Development
AGCOT	Agricultural Growth Corridors of Tanzania
AWF	African Wildlife Foundation
CEOrt	CEO Roundtable of Tanzania
EC	Electrical conductivity
ESARO	Eastern and Southern Africa Regional Office
FFPO	Forest and Farm Producer Organisation
GIS	Geographic Information System
IUCN	International Union for Conservation of Nature
ILM	Integrated Landscape Management
LRI	Landscape Resilience Index
MEL	Monitoring, Evaluation and Learning
MRV	Monitoring, Reporting and Verification
NbS	Nature-based Solutions
NDCs	Nationally Determined Contributions
NGO	Non-Governmental Organisation
Norad	Norwegian Agency for Development Cooperation
NRM	Natural Resource Management
pH	Potential of Hydrogen
PPM	Parts Per Million
PPI	Poverty Probability Index
RESET	Resilient Soybean for Empowerment in Tanzania
RESOLVE-NbS	Scaling-up Nature-based Solutions for Climate Change Resilience in Tanzania
SAGCOT	Southern Agricultural Growth Corridor of Tanzania
SDGs	Sustainable Development Goals
Sida	Swedish International Development Cooperation Agency
SLM	Sustainable Land Management
SNV	Netherlands Development Organisation
SUSTAIN-Africa	Sustainability and Inclusion Strategy for Growth Corridors in Africa
SUSTAIN Eco	SUSTAIN Ecosystem Stewardship to Balance Sustainability and Growth in Tanzania
SUSTAIN Pro	SUSTAIN Productive Landscapes for Inclusive Growth in Tanzania and Mozambique
WUA	Water User Association
WWT	Water and Wetlands Team

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A quick-start guide

The Landscape Resilience Index (LRI) is a practical framework for assessing landscape resilience across three connected pillars: land, water, and adaptive capacity. Developed through the SUSTAIN-Africa initiative, the LRI responds to the need for a flexible monitoring tool that can bring together ecological, hydrological, and socio-economic information in a way that supports decision making across different scales.

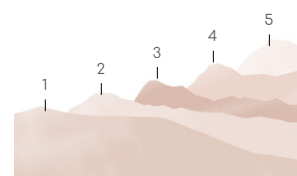
This handbook introduces the structure of the LRI, explains its indicators and scoring approach, and provides step-by-step guidance for applying the framework in a new landscape. It also shows how LRI results can be used to establish a resilience baseline, identify priority areas for action, support integrated landscape management, and inform policy, investment, and adaptive management processes.

Chapter

3

Structure of the LRI:

Three pillars – land, water, and adaptive capacity – defining a landscape's ability to thrive under stress; three scales – site, landscape and national – encompassing geographic scale and governance processes; and 17 indicators, designed to be adaptable across diverse landscape and data contexts. [PAGE ► 6](#)



4

Step by step LRI Implementation:

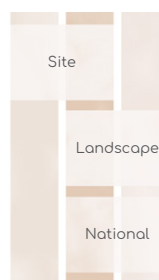
5 steps that can be used by practitioners in a new landscape, including (i) setting up the scope of work, (ii) policy review, (iii) stakeholder engagement, (iv) data collection, and (v) analysis and validation to establish an LRI baseline. [PAGE ► 15](#)



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LRI examples:

Examples of LRI baselines in 2 landscapes of the SUSTAIN-Africa initiative: Barúe-Vanduzi in Mozambique and Kilombero in Tanzania. [PAGE ► 20](#)



1

Introduction

This handbook provides an introduction to the Landscape Resilience Index (LRI), a framework designed to assess current and expected resilience within socio-ecological systems to improve multi-scale decision making.

With landscapes subject to intensifying environmental pressures around the world, resilience is a crucial measure of landscape health and is vital for sustaining ecosystem services, livelihoods, and long-term development [1]. Yet, there is limited understanding about what makes a landscape resilient (see Box 1), due to

fragmented or missing data, a lack of monitoring, disconnects between data and decision making, and limited capacity or funding [3].

The LRI offers a structured, evidence-based approach for integrating diverse indicators to help guide investment, policy decisions, and long-term planning. By translating technical data into a coherent resilience profile, the LRI supports the prioritisation of nature-based solutions (NbS) and integrated landscape management (ILM) approaches to build resilience in economies, nature, and people (see Box 2).

Box 1: Key Definitions

What is a landscape?

A socio-ecological system that consists of natural and/or human-modified ecosystems, and which is influenced by distinct ecological, historical, economic and socio-cultural processes and activities [2].

What makes a landscape resilient?

The capacity to absorb disturbances, adapt to change, and continue to provide essential ecosystem services that sustain human well-being [1].

Box 2: Integrated Approaches

What is ILM?

Integrated Landscape Management (ILM) is a collaborative, multi-stakeholder process aimed at reconciling competing social, economic, and environmental objectives to achieve sustainable landscapes [4].

What is NbS?

Nature-based Solutions are actions to protect, sustainably manage, and restore ecosystems that address societal challenges adaptively while providing human well-being and biodiversity benefits [5].

¹ While the 2017 definition of ILM emphasises achieving “sustainable landscapes,” the LRI framework shifts the terminology to “resilient landscapes” to better capture the dynamic capacity of social-ecological systems to absorb disturbances, adapt to change, and continue to provide essential services amidst intensifying environmental and climate pressures. This shift reflects a move beyond merely “maintaining” resources to focusing on the capacity for renewal and innovation, which is vital for long-term landscape health in the face of increasing global uncertainties.

2



Closing the landscape resilience gap

2.1 The gap in resilience frameworks

Strengthening landscape resilience is essential for reducing vulnerability to climate change, maintaining and enhancing ecosystems and their services, and supporting long-term economic and social development [1]. Yet, practitioners face a significant “resilience gap” due to the limitations of existing frameworks. Approximately 30% of people in tropical countries depend directly on nature to meet their basic needs [6]. These needs are increasingly undermined by land and water scarcity and biodiversity loss driven by increased agricultural land expansion, energy production, and urban sprawl in many regions [7].

A siloed approach to resilience

Many existing tools focus only on individual components of resilience, such as specific ecological or social responses to stressors, or a particular geography, species, or scale, rather than providing integrated assessments across ecological, hydrological, and socio-economic domains [3], see Table 1.

The missing water link

The lack of hydrological indicators within many resilience frameworks is particularly concerning, given that more than two billion people currently lack access to safe drinking water and projections suggest a potential 40% deficit in global freshwater supply by 2030 [11]. Hydrological dynamics such as water availability, storage, and catchment-scale flows are fundamental determinants of landscape resilience [12].

Data and design rigidities

Resilience frameworks are also often too rigid or data-intensive for practical use in diverse landscapes, or those where data, capacity, and funding are limited [13, 14]. As adaptation is a defining characteristic of landscape resilience, frameworks designed to monitor that resilience must also be flexible [15]. Focusing only on quantitative methods or only on participatory and perception-based approaches will limit the content and applicability of the evidence collected. Effective landscape resilience frameworks should be able to integrate diverse datasets, balancing scientific rigour with adaptability to different contexts [16, 17].

The cost of inaction

Finally, many existing tools do not explicitly show how local land and water management actions contribute to progress against broader landscape-scale, national, and international environmental and development policies and goals [18, 19]. Making this link is important for catalysing large-scale investment and

collaboration from both public and private actors, who increasingly seek evidence that interventions align with national policy targets and deliver measurable contributions to climate adaptation, biodiversity recovery, and sustainable development goals.

Table 1: The limitations of existing resilience tools²

Tool	Ecosystem Resilience Index [8]	Spatial Resilience Index [9]	Land Health Monitoring Framework [10]
Focus	Ecosystem structure and function, particularly forest resilience to wildfire	Biodiversity persistence under climate change using spatial modelling	Below-ground diversity, above-ground diversity, and habitat diversity
Limitations	Does not incorporate socio-economic dimensions	Centred only on ecological responses	Limited consideration of hydrological and socio-economic dimensions
Connection to LRI pillars*			 
Indicator rating  Land  Water  Adaptive Capacity *For more information about the 3 pillars, see page 7			

² This table provides a representative sample of existing resilience tools and frameworks analysed during the development of the LRI.

2.2 Building a more complete picture: Why the LRI was developed

The Landscape Resilience Index (LRI) was developed within the SUSTAIN-Africa initiative to overcome these challenges and create a more flexible, integrated approach to monitoring that could be adapted to different scales, varying levels of data availability and institutional capacity.

Integration and connectivity across scales

The LRI captures ecological conditions, hydrological dynamics, and socio-economic dimensions within a single monitoring tool. This allows for a more holistic understanding of how resilience is shaped across landscapes, how land management interventions can generate both synergies and trade-offs over time, and how local actions can be linked to national and international development, biodiversity, and climate goals.

Actionable intelligence to support decision making

The LRI integrates a set of diverse indicators to guide the prioritisation of ILM and NbS, allowing decision makers to balance environmental, social, and economic objectives within a landscape. The LRI is adaptable to diverse data contexts, broadening fixed indicators into overarching themes that can incorporate both primary field data (e.g., bird richness or soil chemistry) and secondary data (e.g., satellite-derived tree cover loss).

A support tool for investment

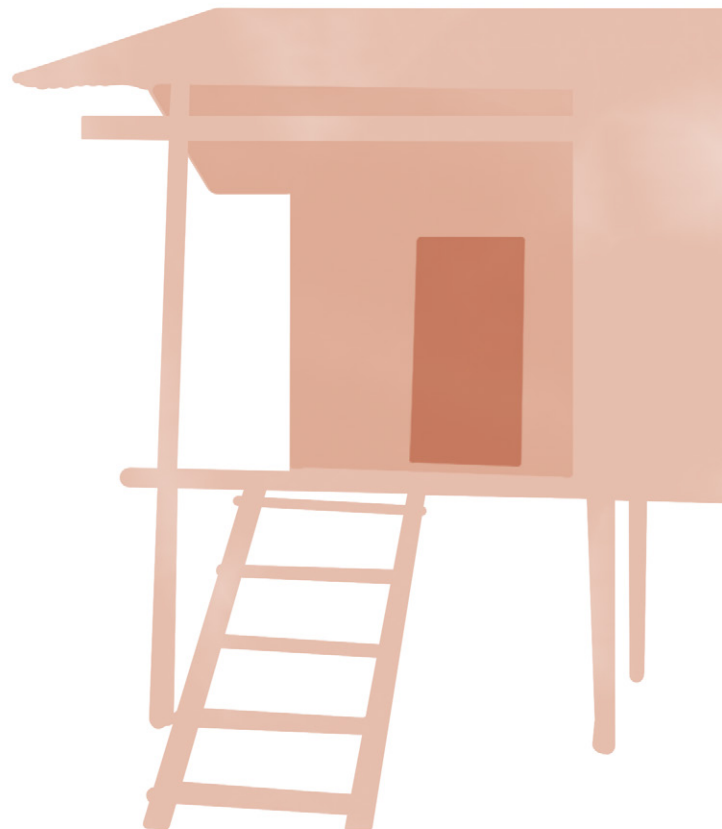
The ultimate rationale for the LRI is to translate technical findings into investable outcomes. By identifying resilience “hotspots,” the LRI helps stakeholders build a compelling evidence base for nature-based solutions that can attract both public and private capital.

3

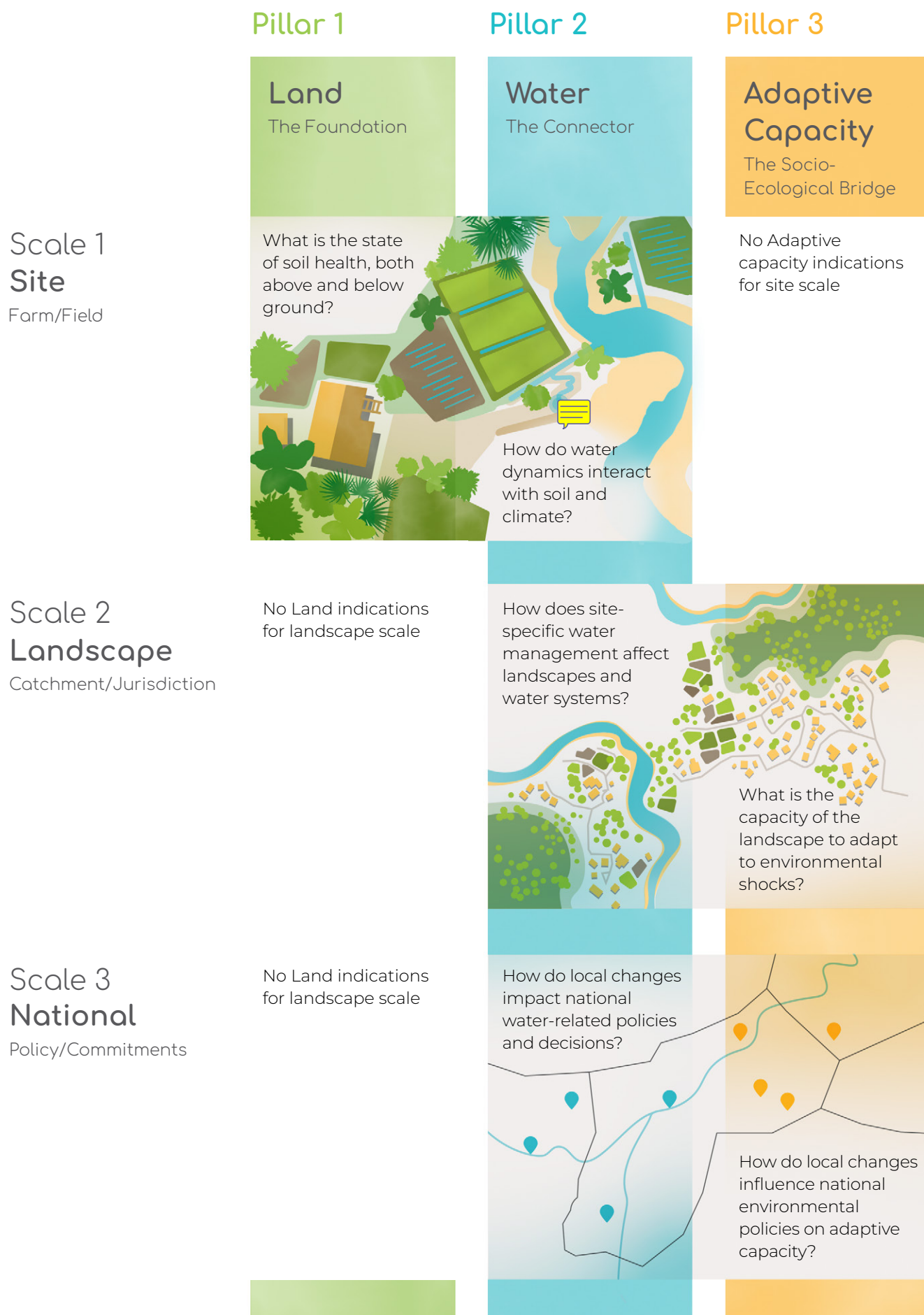
Structure of the LRI

The LRI provides a coherent, holistic picture of landscape health by integrating indicators that were once fragmented across different sectors. It moves beyond traditional ecological monitoring tools by placing hydrological dynamics and human and ecosystem adaptive capacity at the centre of the framework.

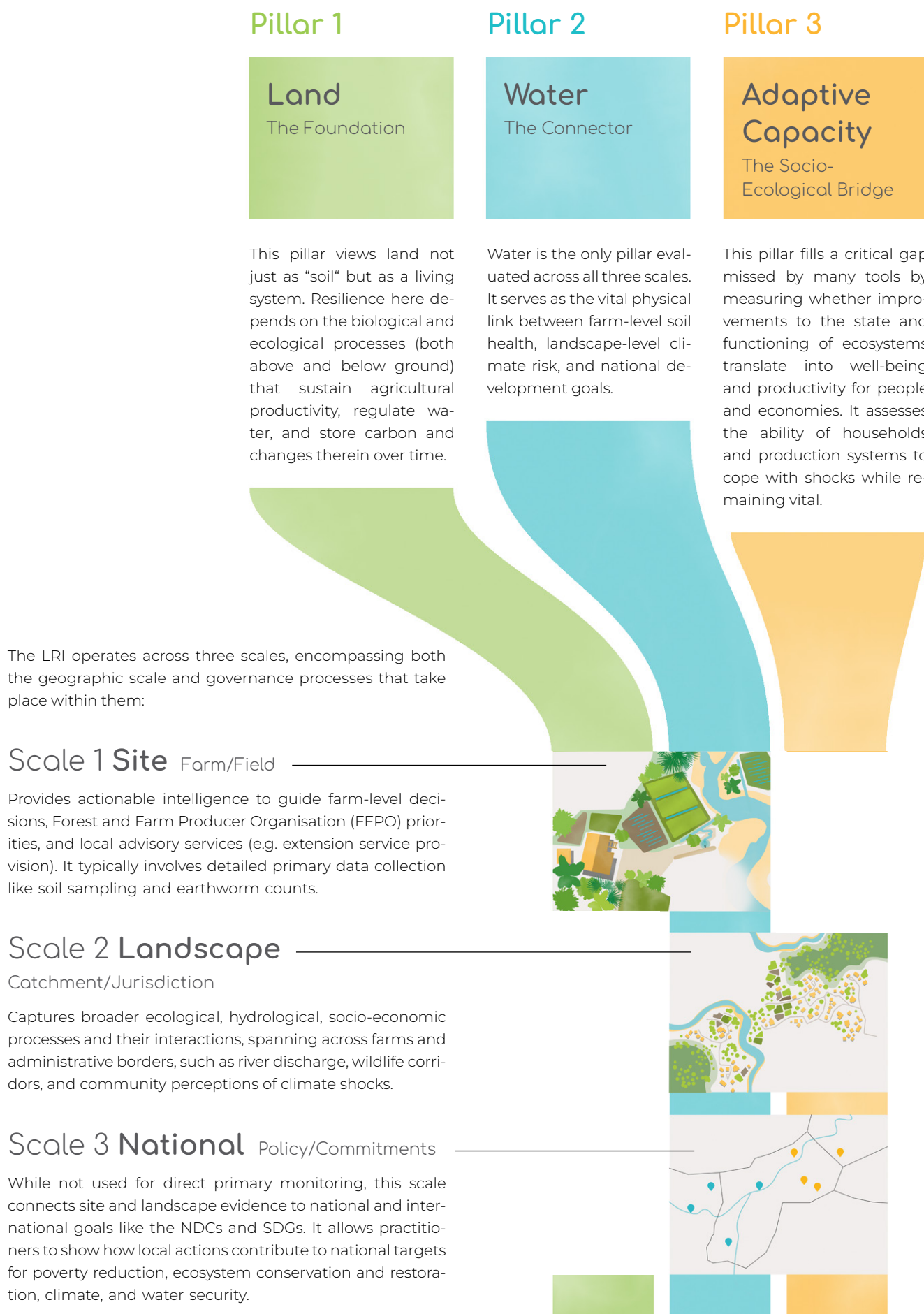
The LRI is organised in three pillars that together define a landscape's ability to persist and thrive under stress. The LRI then operates across three scales, encompassing both the geographic scale and governance processes which take place within those scales. It is composed of a set of 17 indicators, designed to be adaptable across diverse landscape and data contexts. Finally, the LRI as an index has been designed to provide a dynamic assessment of progress towards resilience across its indicators and pillars, rather than a static, one-off measure of resilience.



3.1 A snapshot of the Landscape Resilience Index (LRI)



3.2 The LRI's three pillars and three scales



3.3 The LRI's thematic indicators

Site

Land



Above-ground biodiversity



Below-ground biodiversity



Carbon stocks



Soil fertility

Water



Soil-water availability



Site-level water biological activity



Site-level water chemistry

Adaptive Capacity

Landscape



Landscape water access



Landscape water availability



Landscape water chemistry



Adoption of sustainable land management (SLM) practices



Agricultural, agroforestry and tree plantation productivity



Extreme weather



Household poverty



Landcover percentage



Landscape biodiversity



Perception of shocks and stresses

National



Development of the LRI began with a list of 110 indicators, covering multiple facets of land, water, and adaptive capacity. From that list, the team used several different strength measures, in two separate stages, to narrow down the list.

First, strength measures related to feasibility and scalability were applied, including:

- Costs and the level of technical expertise required;
- Evidence base/availability of data;
- Precision of available data; and
- Scalability across site, landscape, and national levels.

Using these strength measures, and their own expertise and knowledge, the team narrowed the list down to just 20 indicators. The next stage involved applying strength measures related to:

- Relevance to ongoing work within the SUSTAIN-Africa initiative;
- The number of related ecosystem services (provisioning, regulating, cultural, and supportive); and

- The number of related co-benefits, or distinct ecosystem service outcomes.

Along with these three additional strength measures, the team interviewed several stakeholders at the landscape and national level to understand their views and needs. Using these results, as well as feedback from a validation and training workshop and IUCN staff, the final list of 17 indicators was agreed upon.

A defining feature of the LRI is its shift from fixed units of measurement to broad overarching themes. This allows the framework to be adaptable to data-scarce settings and diverse landscape contexts. Practitioners can select sub-indicators within these themes based on local data availability, costs, and technical capacity. For example, “Above-ground biodiversity” might be measured using only bird richness in one landscape, while another landscape might include mammals and invertebrates.

The following pages provide more detail about each of the three pillars.

Land indicators

Site scale



1. Above-ground biodiversity
bird, mammal, and invertebrate species richness



2. Below-ground biodiversity
earthworm count

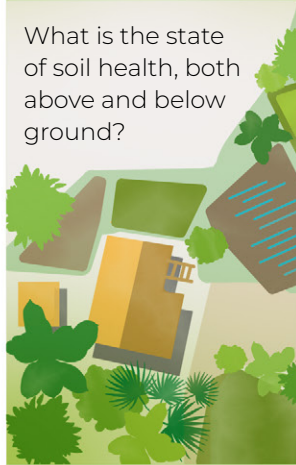


3. Carbon stocks
above-ground and below-ground carbon



4. Soil fertility
electrical conductivity, pH, total nitrogen, soil phosphorous, soil C:N ratio

What is the state of soil health, both above and below ground?



Pillar 1: Land

What is the condition and function of both below- and above-ground biodiversity and ecological processes in the soil of the given site?

Land health is fundamental to resilient agricultural production, ecosystem functioning, and long-term food security. This pillar takes a broader view of land as a living system, by examining both above- and below-ground biodiversity, as well as carbon stocks and soil fertility. This reflects the reality that land resilience depends not only on crop performance, but also on the biological and ecological processes that sustain productivity over time, regulate water, store carbon, and respond to climate and land-use pressures. This framing connects farm-level health with broader landscape restoration, ecosystem services, and sustainable livelihoods, making the LRI more adaptable and policy-relevant.

The **land pillar** of the LRI is evaluated only at the **site scale**, focused on soil and farms.

At this scale, the LRI measures soil biological structures, diversity of crops and habitats, and carbon and fertility levels to determine the health and potential resilience of the land moving forward. Assessing land indicators relies on a combination of primary and secondary data, depending on data availability. It typically involves on-the-ground data collection and lab-based analysis, to generate information that can guide farm-level decisions and advisory services.

Pillar 2: Water

What is the availability, reliability, and quality of water resources in the site and landscape?

Water is both a foundation of resilient landscapes and a key link between land health, climate risk, and livelihoods. The LRI differs from many other resilience frameworks by placing water at the centre of analysis and monitoring it across multiple scales. This framing recognises that resilience depends not only on having water, but on having reliable and sufficient water of appropriate quality, and on understanding how land use and management decisions affect hydrological systems over time. LRI users are encouraged to consider both current water conditions and the wider impact of their activities on water quality, flow, and availability within and beyond their immediate landscape.

Water is the only one of the three LRI pillars that is evaluated at all three scales – **site, landscape, and national**.

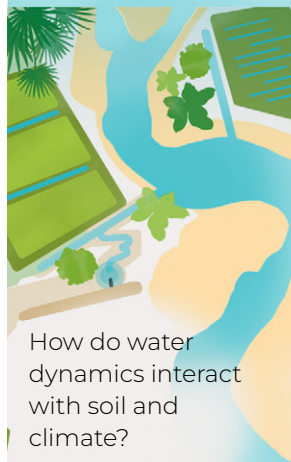
At the site scale, the LRI measures availability of water in soil, as well as water quality and chemistry at the level of individual farms or sites, to guide farm-level decisions, FFPO priorities, and advisory services.

At the landscape scale, the LRI captures broader ecological and hydrological processes that span farm and administrative borders, such as catchment-level water access, availability, and quality. As with land, measurement for most water indicators relies on on-the-ground data collection and lab-based analysis of primary and secondary data, although landscape water access and availability rely on secondary data only.

At the national scale, the LRI connects local water monitoring efforts with national policies and international goals, which can improve decision making and increase understanding of how actions in one landscape can impact resilience elsewhere. Among the national scale indicators that LRI results can feed into are:

- National-level water quality and quantity (both water and wastewater)
- Access to quality water for different uses at the national level (including sanitation and irrigation).

Water indicators



How do water dynamics interact with soil and climate?



How does site-specific water management affect landscapes and water systems?



How do local changes impact national water-related policies and decisions?

Site scale



5. Soil-water availability
soil moisture, plant-available water content, volumetric water content



6. Water quality: small-stream biological activity
macroinvertebrate assemblages



7. Site-level water chemistry
pH, electrical conductivity, total dissolved solids

Landscape scale



8. Landscape water access
household consumption, irrigation



9. Landscape water availability
river discharge



10. Landscape-level water chemistry
pH, electrical conductivity, total dissolved solids

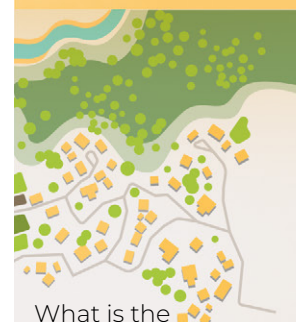
National scale



Water quality and quantity
water and wastewater



Access to quality water for different uses
drinking, sanitation, irrigation



What is the capacity of the landscape to adapt to environmental shocks?

Pillar 3: Adaptive capacity

Are people and systems able to adapt to climate change and related environmental shocks while maintaining their well-being and productivity?

Resilience cannot be understood only in ecological terms; it must also capture whether people, production systems, and local economies are able to cope with shocks and adapt over time while maintaining well-being and productivity. By bringing together social, economic, climate-risk, and land-management dimensions, this pillar fills an important gap in many existing frameworks. Evaluating adaptive capacity can illuminate whether improvements in land and water management translate into stronger resilience for households and landscapes. It also makes the LRI more relevant for national policy processes, by connecting local experience to wider priorities on climate adaptation, poverty reduction, food security, and resilient rural development.

The **adaptive capacity pillar** is evaluated both at the **landscape and national scales**.

At the landscape scale, it looks at ecological connectivity, use of sustainable land management practices, exposure to extreme weather events, and socioeconomic conditions. Here, individual actions are linked to broader environmental outcomes, which is essential for landscape management, ecosystem restoration, and climate resilience. Unlike many of the site and landscape indicators in the land and water pillars, all of the landscape scale indicators for adaptive capacity rely solely on secondary data.

Landscape scale



11. Adoption of Sustainable Land Management (SLM) practices

soil fertility, soil erosion, use of manure, crop rotation, use of compost, agroforestry



12. Agricultural, agroforestry and tree plantation productivity

crop yield



13. Extreme weather

drought, flooding



14. Household poverty

poverty probability index, insufficient income to buy food



15. Land cover percentage

tree cover and loss



16. Landscape biodiversity

biodiversity habitat index



17. Perception of shocks and stresses

drought, flooding

At the national scale, adaptive capacity results from the LRI can be integrated into national reporting systems and data collection efforts, to support decision making and action on national and international policies and goals. Among the national-scale indicators that LRI results can feed into are:

- Agricultural productivity, when the LRI assessment includes indicators on nationally relevant crops;
- Carbon stored in forest and agricultural land, including associated emissions reductions;
- Percentage of national land impacted by adoption of Sustainable Land Management (SLM) practices, certification, and agroecological markets;
- Monitoring of healthy and diverse ecosystems on both natural and restored land;
- The level of community resilience to shocks, disasters, and climate change; and
- The national poverty rate, based on however the country measures poverty.

See Annex A, showing the example sub-indicators used under each LRI pillar across the four SUSTAIN-Africa landscapes.

See Annex B, for more information on LRI's scoring thresholds.

National scale



Agricultural productivity



Carbon stored in forest and agricultural land



Adoption of SLM practices and agroecological markets



Healthy and diverse ecosystems



Resilient communities



Poverty rate



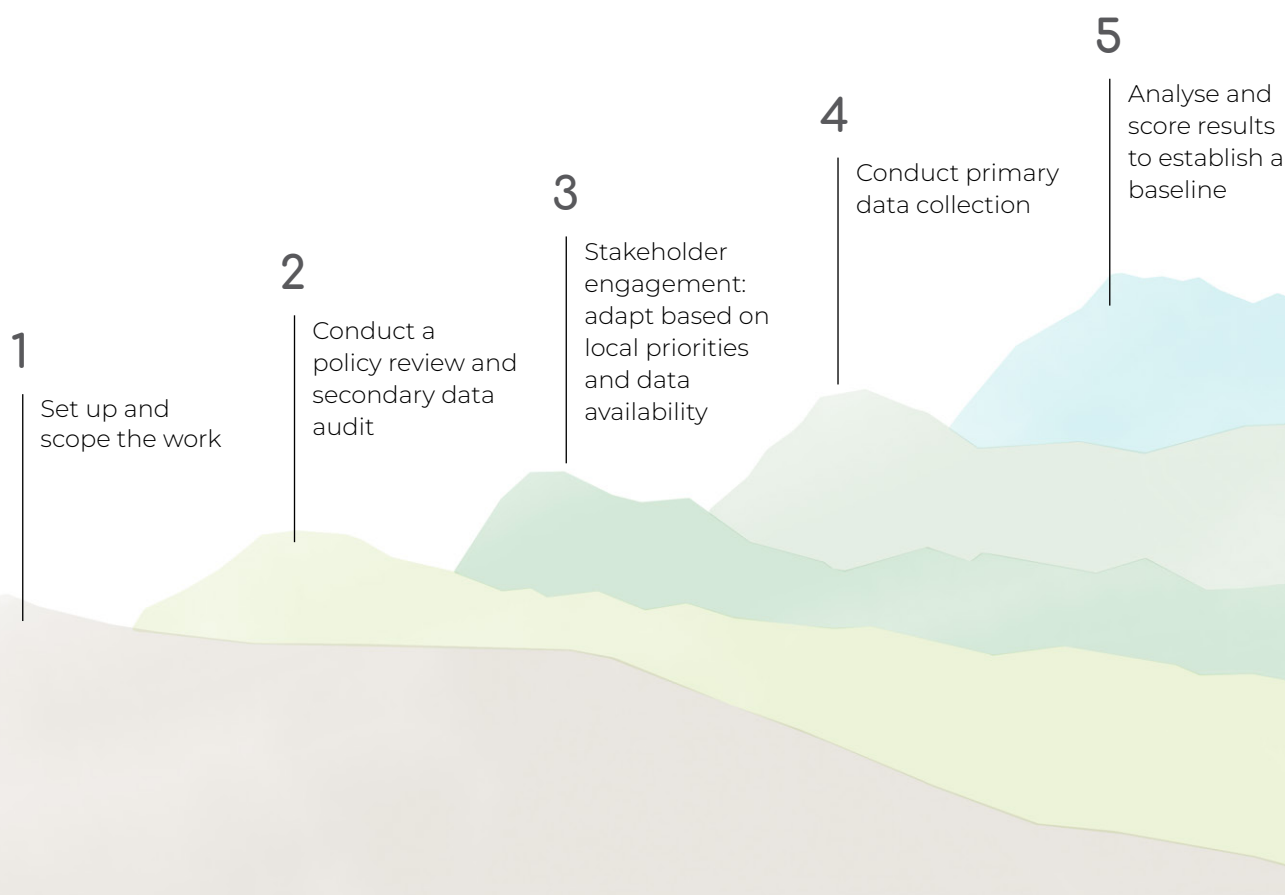
How do local changes influence national environmental policies on adaptive capacity?

4



How to implement the LRI in a new landscape

While the LRI was developed and piloted as part of the SUSTAIN-Africa initiative in Mozambique and Tanzania, it is designed to be applied globally, in any landscape. This section presents a series of steps that can be used by practitioners seeking to implement the LRI in a new landscape (see the 5 steps below).



Step 1: Set up and scope the work

Before any data collection begins, the first step involves defining the scope of the work and establishing the team, including NGOs, data coordinators, GIS experts, field teams, and technical specialists, who will implement the LRI analysis. The team should begin by defining the boundary of the landscape to be assessed, identifying communities within that landscape, and mapping key actors, such as farmers, cooperatives, water users, district authorities, NGOs, and government agencies. It is also helpful at this point to clarify what the LRI assessment will be used for, for example to prioritise interventions, for monitoring or policy reporting, or to build an investment case for ILM and NbS.

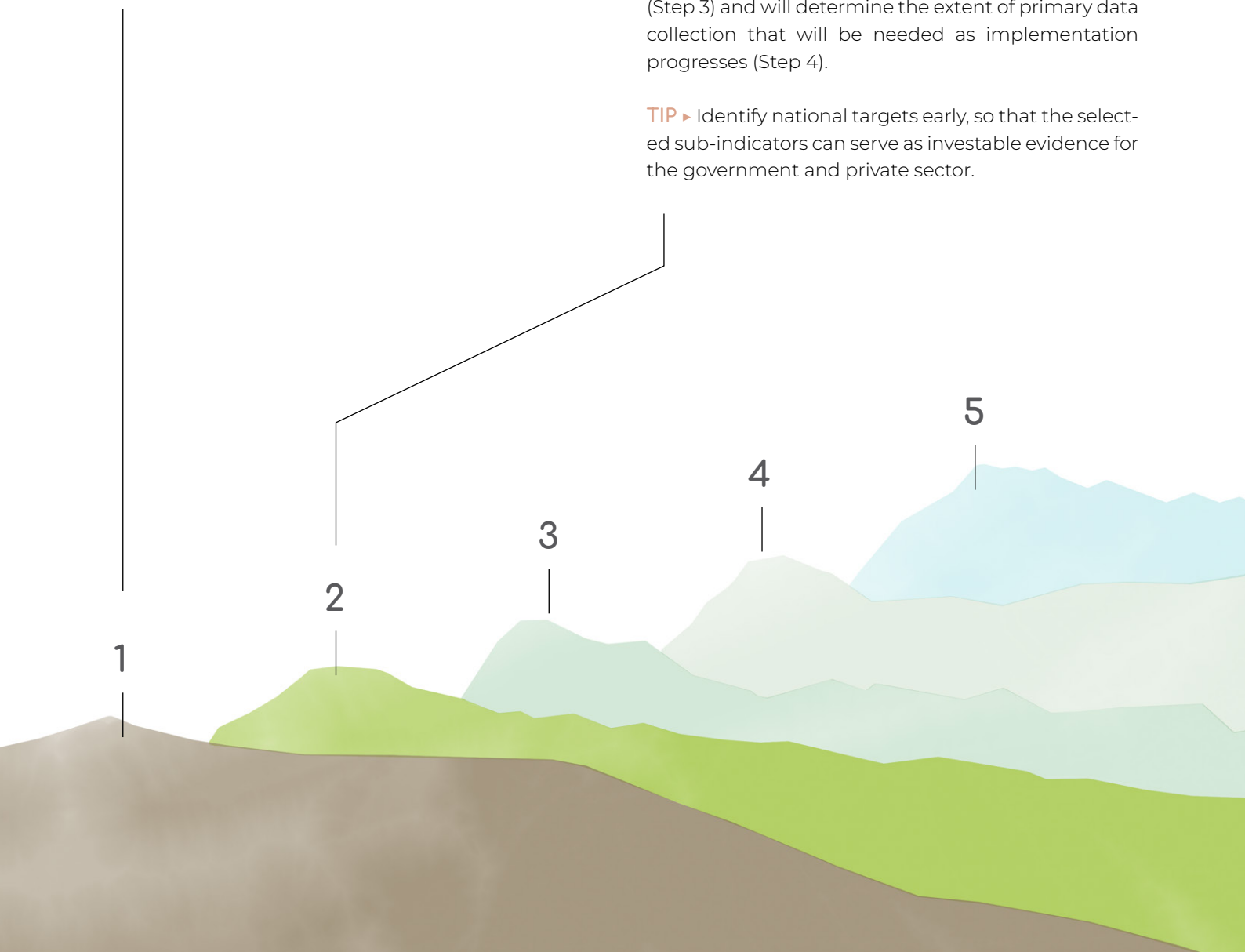
TIP ► Delineate boundaries based on socio-ecological systems (e.g., watersheds or wildlife corridors) rather than just administrative lines, to ensure hydrological processes are captured.

Step 2: Conduct a policy review and secondary data audit

Next, the team should conduct a policy review, at regional and national scales, to identify and establish the national priorities and government commitments relevant to the LRI. This will help when it comes time to feed water and adaptive capacity results from the LRI into national-scale indicators.

The team should also review any existing soil, water, biodiversity, climate, land-cover, or livelihood data that fits within the scope of the LRI indicators. This audit can help guide the choice of sub-indicators (Step 3) and will determine the extent of primary data collection that will be needed as implementation progresses (Step 4).

TIP ► Identify national targets early, so that the selected sub-indicators can serve as investable evidence for the government and private sector.



Step 3: Stakeholder engagement

At this point in implementation of the LRI, it is important to involve key landscape actors (as identified in Step 1) to more fully develop an implementation plan. This is a crucial step for adapting the broad framework of the LRI to the particular situation, capacity, and priorities of the landscape. A stakeholder workshop is a good way to bring key actors together to discuss existing capacity in the landscape, available data, and opportunities for cooperation and collaboration.

Using the three LRI pillars (land, water, adaptive capacity), the participants should review the data collected during the Step 2 audit to determine which sub-indicators will be relevant to the local landscape. Which sub-indicators to focus on will depend on existing capacity challenges and opportunities, for example the costs and level of technical expertise required, the availability and precision of data, and scalability across site, landscape, and national scales. The team should also consider the relevance of the sub-indicators to current activities and ecosystem services in the landscape.

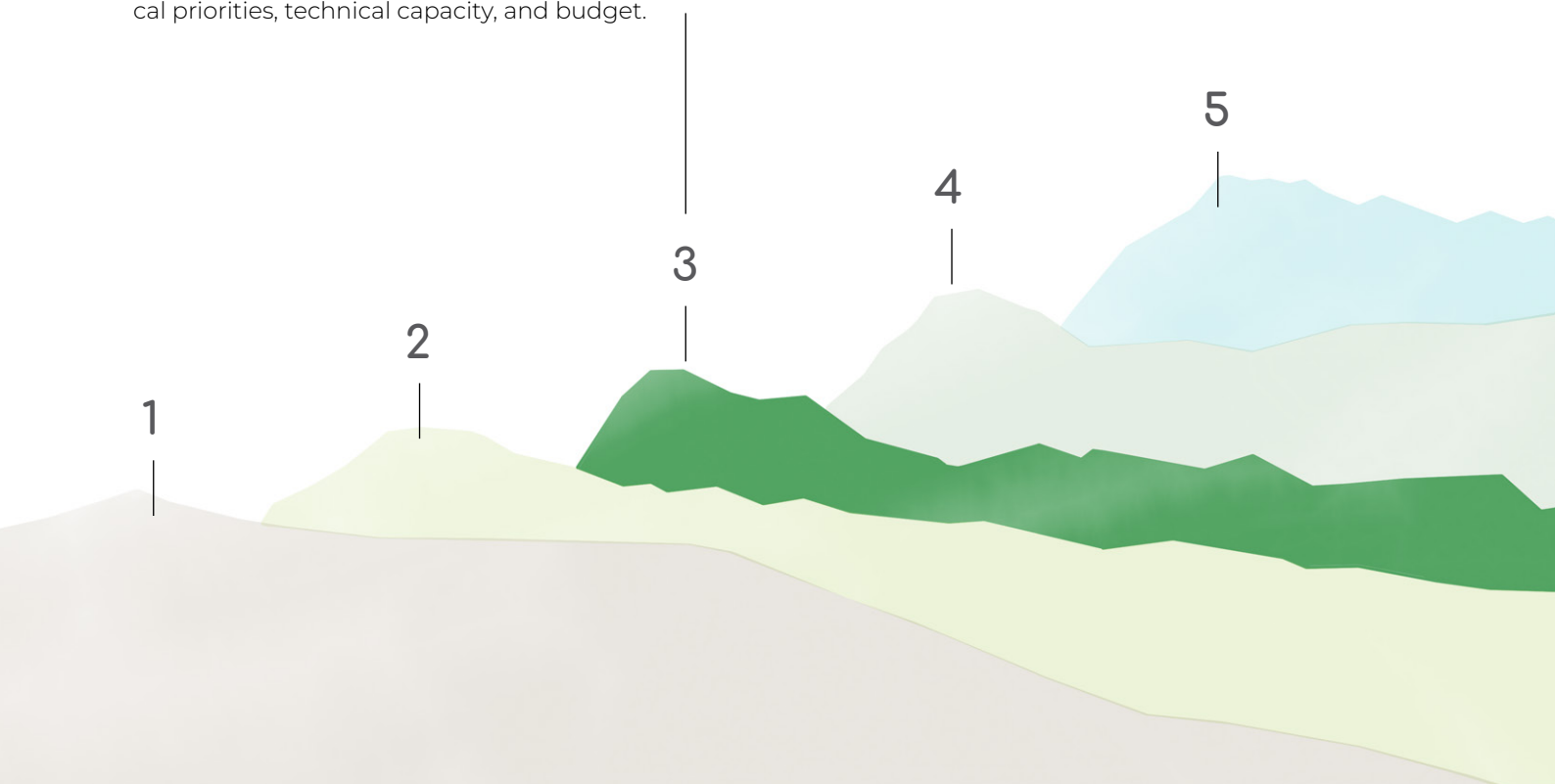
- The stakeholder engagement process should result in a set of relevant sub-indicators for the LRI team to focus on (see Box 3).
- See Annex A for examples of sub-indicators used under each LRI pillar across the four SUSTAIN-African landscapes.

TIP ▶ Instead of mandating all sub-indicators, allow stakeholders to make the final selection based on local priorities, technical capacity, and budget.

Box 3: Indicators and sub-indicators

While the LRI lays out a set of 17 indicators across the three pillars to guide data collection, the choice of sub-indicators within each of those indicators will vary depending on the particular landscape context. For example, in the four SUSTAIN-Africa landscapes:

- The site-scale land indicator above-ground biodiversity was measured just by bird-species richness in some landscapes and by bird-, invertebrate-, and mammal-species richness in others.
- The site-scale water indicator soil-water availability was measured variously as plant available water content, water holding capacity, or soil moisture.
- The landscape-scale water indicator landscape water access was measured as access to water for home consumption or percentage of people reporting access problems.
- The landscape-scale adaptive capacity indicator agricultural agroforestry and tree plantation productivity was measured based on a wide range of different crops, including rice, sunflower, soybean, sugar cane, beans, maize, pigeon pea, green bean, cabbage, and potato.



Step 4: Conduct primary data collection

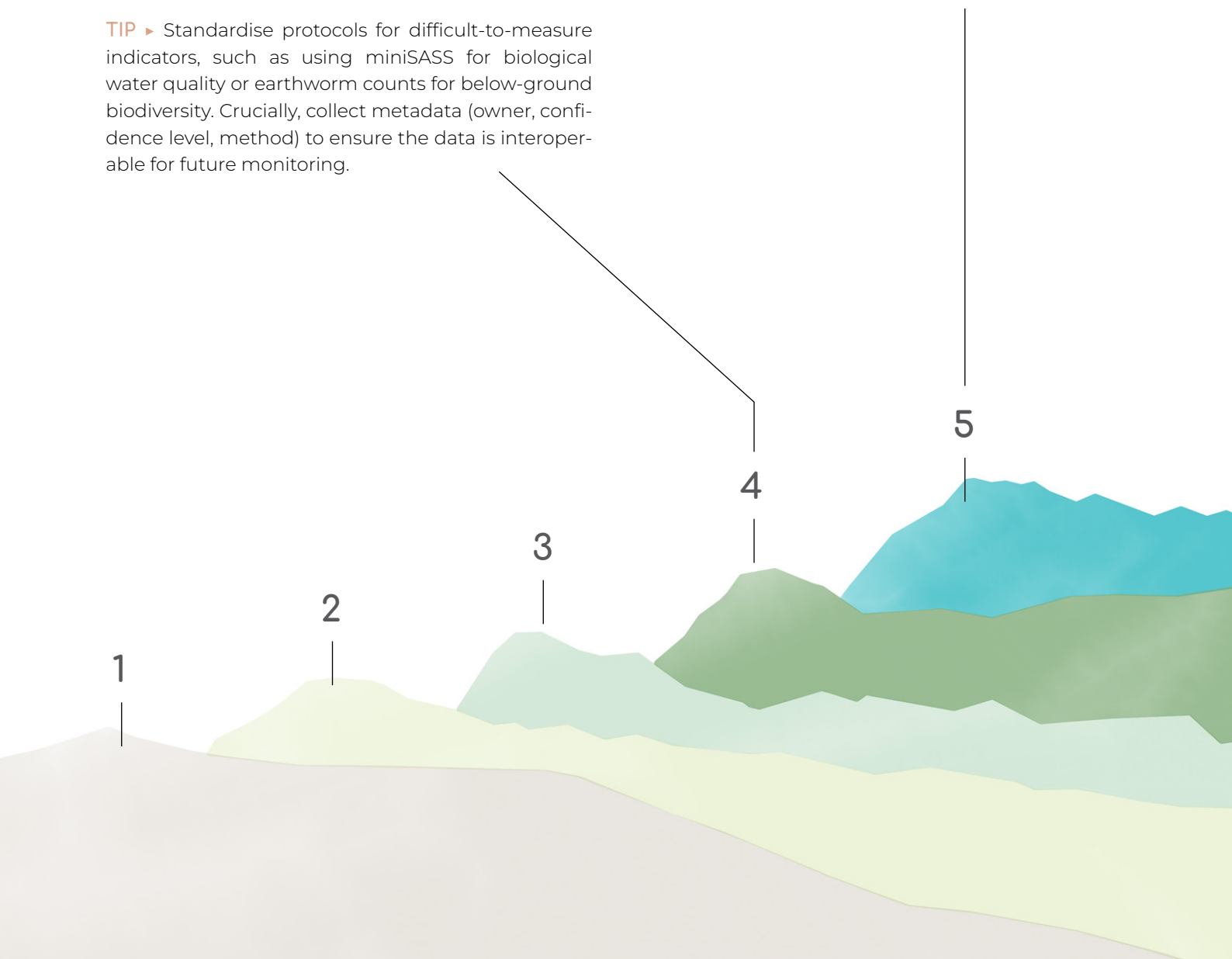
This step is the core of the LRI, when the team collects data that will make up the landscape's resilience baseline. Sampling should be organised by land use, catchment, intervention area, and season, if possible. Permits should be obtained, and protocols, equipment, and lab arrangements prepared. This is also the time to conduct any additional training needed for data collectors and partners. When taking samples and collecting data, the team should also be sure to collect metadata, including location, date, season, method, data owner, and confidence. This will help ensure that the data can be replicated, monitored, and verified.

TIP ► Standardise protocols for difficult-to-measure indicators, such as using miniSASS for biological water quality or earthworm counts for below-ground biodiversity. Crucially, collect metadata (owner, confidence level, method) to ensure the data is interoperable for future monitoring.

Step 5: Analyse and score results to establish a baseline

Once all data has been collected, cleaned up, and standardised, the team can score the sub-indicators according to established thresholds or standards, if they exist, and expert judgement if they do not. The collected and aggregated land, water, and adaptive-capacity data from these indicators will make up a baseline resilience profile of the landscape across site, landscape, and national scales. This will reveal more clearly where resilience challenges are most pronounced, where gaps in data might still exist, and where improvements to land management could deliver the greatest environmental, social, and economic benefits.

TIP ► Apply the four-level scoring system. Where scientific literature or policy standards are missing, use a formalised expert judgement process to set provisional thresholds, noting that these are adaptive and should be reviewed every three years.



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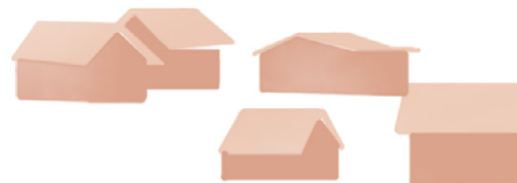
Pathways for using the LRI

Once the five steps in Chapter 4 have been completed and a landscape resilience baseline profile is available, the next phase is to translate these results into actionable intelligence for public institutions, private actors, and development partners to prioritise landscape interventions that strengthen resilience by contributing to climate adaptation, biodiversity recovery, food security, and sustainable development objectives.

While this phase is largely beyond the scope of this handbook, the section provides a brief summary of three avenues for applying the LRI to directly support planning, prioritisation, and investment decisions.

The LRI can support activities and decision-making by:

- Establishing a **baseline of evidence on landscape resilience** to identify priority areas for intervention and investment. Stakeholder engagement will be an important part of determining how and where to align the LRI results with national targets. A participatory process at various levels can help with prioritising interventions to align with international land, water, climate, and development policies and commitments. This will help determine where those interventions can deliver the most immediate resilience and livelihood benefits.
- Supporting continued **monitoring, reporting, and verification** to capture long-term trends and intervention impacts and facilitate adaptive management. Once interventions have been identified and implemented, changes in landscape resilience can be monitored and tracked over time through periodic measurements of the indicators. This will help assess whether interventions are improving resilience outcomes, provide evidence for any necessary adaptive management or refining of interventions, strengthen the case for continuing or increased investment, and support the scaling of successful approaches across landscapes and national programmes.
- Providing a mechanism to integrate landscape-scale resilience evidence into **planning, governance, and investment**. This integration can support institutional uptake and long-term scaling of land-use strategies and decisions. By highlighting risks, opportunities, and potential co-benefits within the landscape, LRI indicator results can help guide investment priorities, programme targets, and engagement with the private sector.



LRI baseline profiles

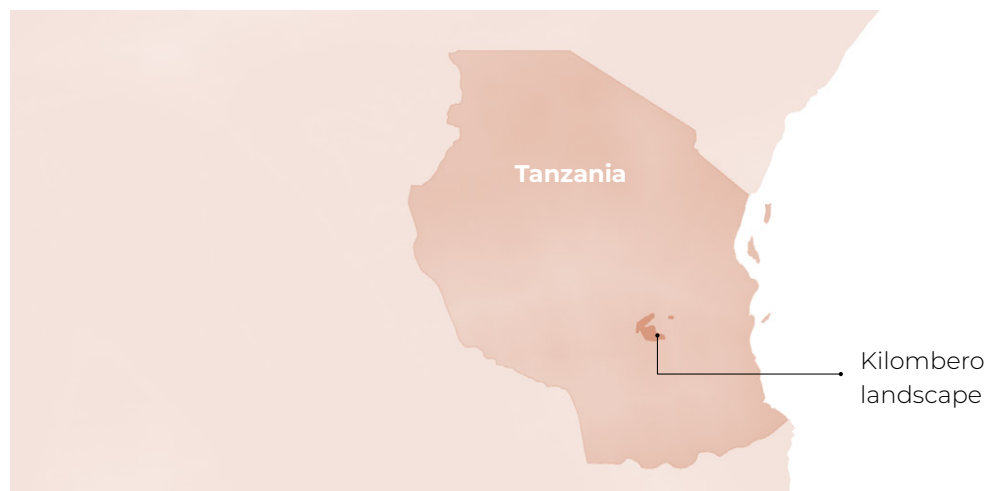
The LRI was applied in four different landscapes as part of the SUSTAIN-Africa initiative: Barué-Vanduzi in Mozambique and Kilombero, Ihemi, and Sumbawanga in Tanzania. These applications were used to test how the LRI can generate a more integrated picture of landscape resilience, identify priority areas for action, and support the use of landscape-level evidence in monitoring, policy³, and investment processes.

The two baseline profiles presented below are not intended to provide a full evaluation of SUSTAIN-Africa interventions. Rather, they illustrate how the LRI can be used to organise diverse data into a practical assessment of landscape resilience. Each profile shows how indicators across land, water and adaptive capacity can be interpreted together to identify strengths, emerging risks, and possible implementation priorities.

³ A series of suggested policy recommendations and priority actions for each country can be found in the IUCN [policy brief](#).

Baseline profile 1:

Applying the LRI in Kilombero, Tanzania



The Kilombero Valley of Tanzania lies between key conservation areas, including the Selous Game Reserve and the Udzungwa Mountains. The area is noted for both high agricultural productivity and biodiversity value. It is home to a significant portion of the world's Puku antelope population and encompasses the Kilombero Valley Floodplain, one of Africa's largest wetlands and a designated Ramsar site. However, agricultural expansion and shifting land use are contributing to ecosystem degradation and water management challenges in the region.

The SUSTAIN-Africa Kilombero landscape covers the districts of Kilombero, Ifakara, Kilosa, Malinyi, and Mlimba within the Southern Agricultural Growth Corridor of Tanzania (SAGCOT). The initiative's efforts in Kilombero focused on implementing climate-smart agricultural practices and improving governance to sustain ecosystem services. In this context, the LRI was applied to develop a baseline profile of landscape resilience and to identify where future implementation, monitoring and investment could be better targeted. The results of this assessment are shown in Table 2.

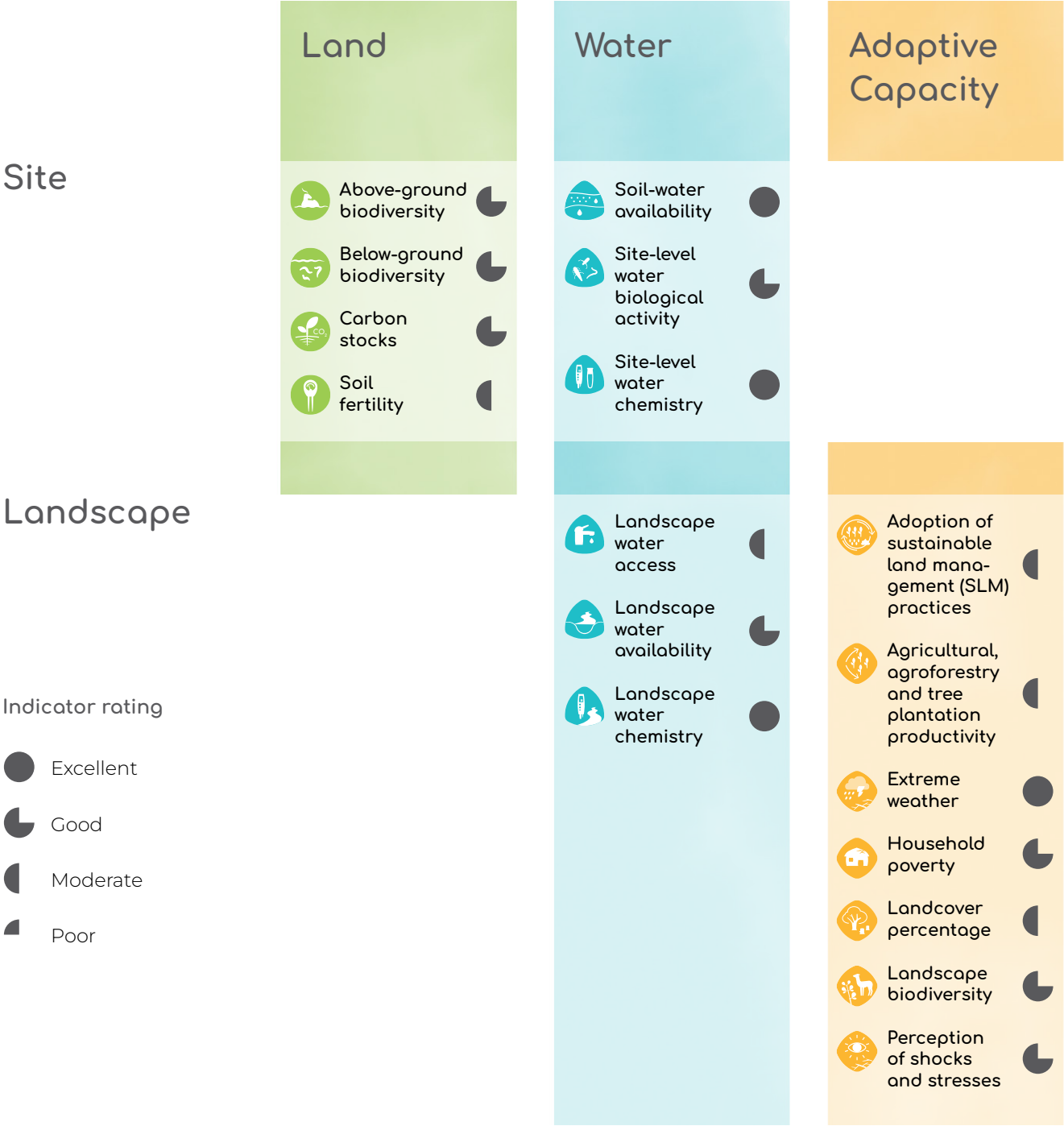
The Kilombero profile illustrates how the LRI can support three linked functions: building a more complete picture of landscape resilience, generating actionable intelligence for implementation, and translating technical findings into evidence that can inform policy and investment decisions.

Interpreting the Kilombero baseline profile

The Kilombero baseline profile highlights good ecological conditions and moderate-to-good resilience across most site- and landscape-scale indicators. Stronger scores were recorded for water availability and quality, carbon stocks, and biodiversity, suggesting that important ecological and hydrological functions remain intact in parts of the landscape. However, the profile also demonstrates areas where the landscape may be vulnerable if pressures continue or intensify. Moderate scores were recorded for soil fertility, landscape water access, adoption of sustainable land management practices, agricultural and agroforestry productivity, and land-cover percentage. These indicators point to a set of linked challenges: maintaining productive agricultural systems, improving soil and water management, reducing land degradation, and ensuring that ecosystem functions are sustained as land-use pressures increase.

The value of the LRI in this context is that it does not treat these issues as separate problems. Instead, it shows how site-level conditions, such as soil fertility and soil-water availability, connect to landscape-level dynamics, such as water access, land cover, biodiversity and adaptive capacity. This helps practitioners identify where integrated landscape management approaches, sustainable land management practices, and nature-based solutions can be combined to strengthen resilience.

Table 2: Landscape resilience baseline profile for the Kilombero landscape⁴



⁴ Primary data collection, including field sampling and laboratory analysis, was focused specifically on villages and target areas where the SUSTAIN initiative is active. While these localised results provide a foundational snapshot of the state of the landscape, they reflect specific land uses and socio-ecological contexts and should be generalised across the broader landscape with caution. To enhance the precision of the resilience profile and account for spatial variability, expanded sampling and continuous monitoring are recommended whenever resources allow.

From baseline findings to implementation direction

Based on these results, the SUSTAIN-Africa team proposed a series of recommendations that would enhance long-term sustainability and contribute directly to Tanzania's national policy and development commitments:

- **Improved access to water for multiple uses:** At the site scale, expanding small-scale irrigation infrastructure would improve crop productivity and reduce sensitivity to rainfall variability and drought, while at the landscape scale, protected wells, irrigation schemes, and managed water access points would improve access to safe drinking water. These interventions would also contribute to Tanzania's national progress under several SDGs and its national climate change response objectives.
- **Sustainable intensification of agriculture:** At the site scale, improved nutrient management, organic amendments, crop rotation, and composting would improve soil fertility, enhance yields, and build resilience to climate stress, while also supporting national agricultural productivity objectives and emissions reduction and carbon storage targets. At the landscape scale, scaling up adoption of sustainable land management and agroecological practices – such as crop rotation, compost application, and practices to reduce soil erosion and enhance soil fertility – would strengthen ecosystem function, stabilise production, and reduce vulnerability to shocks. These practices would also directly contribute to national commitments under the Tanzania National Ecological Organic Agriculture Strategy, the National Climate Change Response Strategy, and other climate-smart management plans and policies.

In this sense, the LRI helps move from indicator results to practical implementation pathways. It shows where risk mitigation is needed, for example through sustainable land management and improved water governance, and where opportunity creation is possible, for example through restoration, agroforestry and nature-based solutions.

Linking the baseline profile to policy and investment

The Kilombero baseline profile also demonstrates how LRI results can support policy alignment and

investment planning. Moderate scores can be used as early signals of where national or landscape-level targets may be difficult to achieve without additional action. For example, moderate performance on soil fertility and sustainable land management is directly relevant to national objectives on land health, ecological organic agriculture, and climate-smart land management.

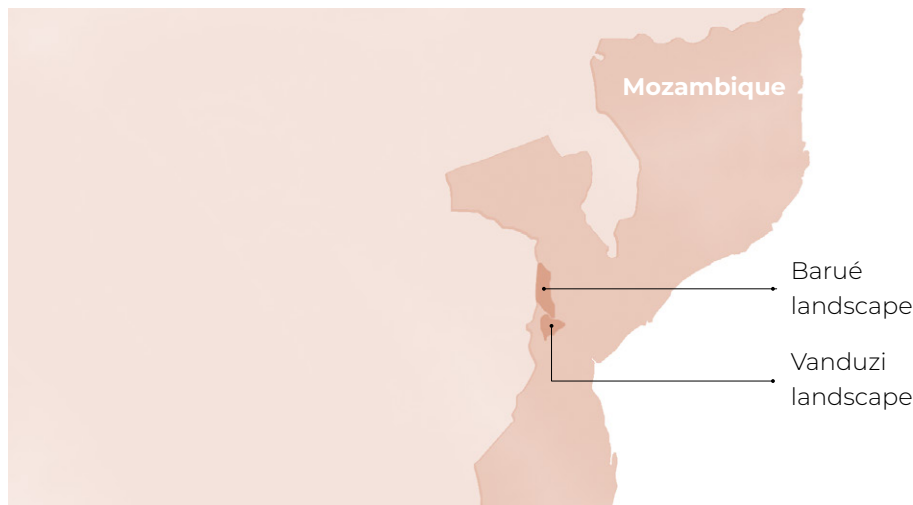
The LRI can therefore help practitioners and decision makers move from monitoring to investment readiness. Rather than designing separate interventions for each moderate indicator, the results can be grouped into integrated landscape investment priorities. For example, improving sustainable land management can also support soil fertility, productivity, water retention, and ecosystem health. This creates a stronger case for coordinated public, private, or grant-based investment in landscape resilience.

Box 4: Future of the LRI in Tanzania

The LRI has evolved into a demand-driven monitoring tool with significant institutional uptake. Its future in Tanzania is defined by its ability to bridge the gap between technical data and high-level policy decisions through the following:

- Evidence generated through LRI baseline profiles contributed to discussions around Tanzania's first Land Health Management Compact, signed in November 2024.
- Work is progressing to institutionalise the LRI as the primary monitoring tool for the Agricultural Growth Corridors of Tanzania (AGCOT) secretariat. This effort is a collaborative initiative between AGCOT, the Vice President's Office (VPO), and the CEO Roundtable (CEOrt) of Tanzania to ensure that landscape resilience indicators are embedded directly into the planning and reporting processes of the country's major growth corridors.
- The LRI is being integrated into other IUCN projects, including RESOLVE-NbS and RESET. These programmes will utilise the index to ensure that interventions in food systems and ecosystem stewardship are evidence-based.

Baseline profile 2: Applying the LRI in Barué-Vanduzi, Mozambique



The Barué-Vanduzi landscape is situated within the Beira Agricultural Growth Corridor of Mozambique, in Manica Province. The landscape has high agricultural potential and strategic importance for food production, with maize, soy, and horticultural crops forming important parts of local production systems. Maize is a key staple crop, soy is important for poultry feed and oil production, and horticultural products offer high-value market opportunities.

The landscape faces a number of pressures that may affect long-term resilience. These include limited access to quality seeds, underdeveloped agricultural extension services, market and processing bottlenecks, rapid population growth, land-use change, and pressure on natural resources.

The SUSTAIN-Africa Barué-Vanduzi landscape covers the districts of Chibuto, Macossa, Nhatsanze, 7-Abril-1, 7-Abril-2, and Nhamandembe. The LRI was applied to develop a baseline assessment of landscape resilience in Barué-Vanduzi. The results of this assessment are shown in Table 3.

Interpreting the Barué-Vanduzi baseline profile

The Barué-Vanduzi baseline profile shows a mixed resilience picture. Several indicators suggest encour-

aging levels of adaptive capacity and water quality, including good scores for soil fertility, adoption of sustainable land management practices, household poverty, landscape biodiversity, and extreme weather, as well as excellent scores for large-river chemical water quality, small-stream chemical properties, and perception of shocks and stressors.

However, the baseline also highlights clear areas of concern. Poor scores for carbon stocks, small-stream biological activity, and land-cover percentage point to pressures on vegetation, habitat quality, stream ecosystem health, and landscape structure. Moderate scores for above-ground and below-ground biodiversity, soil-water availability, landscape water access, landscape water availability and productivity also suggest that the landscape's resilience could weaken over time if land-use pressures, water access constraints, and ecosystem degradation are not addressed.

The value of the LRI in this context is that it brings these signals together into one landscape resilience profile. Rather than treating carbon stocks, water access, biodiversity, land cover, and productivity as separate issues, the LRI shows how they interact as part of a wider socio-ecological system. This helps practitioners identify where integrated landscape management, sustainable land management, and nature-based solutions can reinforce one another.

Table 3: Landscape resilience baseline profile for the Barué-Vanduzi landscape⁴

	Land	Water	Adaptive Capacity
Site	 Above-ground biodiversity   Below-ground biodiversity   Carbon stocks   Soil fertility 	 Soil-water availability   Site-level water biological activity   Site-level water chemistry 	
Landscape		 Landscape water access   Landscape water availability   Landscape water chemistry 	 Adoption of sustainable land management (SLM) practices   Agricultural, agroforestry and tree plantation productivity   Extreme weather   Household poverty   Landcover percentage   Landscape biodiversity   Perception of shocks and stresses 
Indicator rating	 Excellent  Good  Moderate  Poor		

From baseline findings to implementation direction

Based on these results, the SUSTAIN-Africa team proposed a series of recommendations for Barué-Vanduzi that would improve multiple site- and landscape-scale indicators while also contributing to national targets:

- **Reverse tree cover loss:** Nature-based solutions such as riparian restoration, buffer-zone creation, and restoration of degraded areas could reverse deforestation at the site and landscape scales, increasing carbon stocks and enhancing habitat quality and connectivity. At the national scale, promoting agroforestry would contribute to national plans for reducing greenhouse gas emissions and conserving biodiversity.
- **Improve soil carbon stocks:** Interventions such as the application of manure and retention of organic residues could help increase soil organic carbon in croplands, reducing the impacts of drought, enhancing crop yields, and supporting higher soil biological activity. These methods would also contribute to national-scale carbon storage targets.
- **Address water access:** Installing safe, managed water access points and establishing buffer zones along rivers and streams to reduce contamination would increase availability of clean, safe water. In addition, expanding irrigation infrastructure at the site scale would help improve agricultural activity at the landscape scale. The social, health, and safety benefits of such interventions would also contribute toward Mozambique's progress on SDG 6 and its national agricultural plan.

Linking the baseline profile to policy and investment

The Barué-Vanduzi profile illustrates how the moderate and poor scores provide a practical basis for identifying where resilience gaps are most pronounced and where investments could generate multiple benefits. For example, actions to restore tree cover and improve land-cover conditions could support carbon storage, biodiversity, habitat connectivity, and climate

mitigation objectives. Investments in soil health could improve productivity, drought resilience, and soil biological activity. The LRI can therefore help practitioners group related resilience gaps into integrated landscape investment priorities. Rather than developing separate interventions for carbon, water, soil, and productivity, the baseline profile can be used to identify packages of actions that address several indicators at once. This can strengthen the case for blended finance, public investment, grant funding, and private-sector engagement in landscape resilience.

Box 5: Future of the LRI in Mozambique

The future of the LRI in Mozambique is linked to its potential use as both a monitoring tool and an investment-prioritisation tool.

- A collaboration has been initiated with the Mozambique MRV team hosted by the Ministry of Agriculture, Environment and Fisheries to explore how the LRI could support monitoring, reporting, and verification processes. This provides an opportunity to connect landscape-level resilience indicators with wider national monitoring systems and to strengthen the use of land, water, and adaptive-capacity data in policy and planning.
- Blended finance discussions have been initiated with GAPI to explore how the LRI could help prioritise investment decisions. By identifying resilience gaps through the landscape baseline profile, the LRI can help guide investment towards areas where financing could generate multiple benefits, including improved land health, water access, ecosystem restoration, productivity, and livelihood resilience.
- The LRI is also expected to be used in other IUCN projects in Mozambique, where it can support evidence-based planning for food systems, ecosystem stewardship, restoration, and integrated landscape management.



The way forward

The Landscape Resilience Index (LRI) developed through the SUSTAIN-Africa initiative, brings together land, water, and adaptive capacity indicators into a single, usable framework for assessing landscape resilience. Its application in Tanzania and Mozambique has shown how diverse data can be translated into baseline profiles that help identify resilience gaps, guide integrated landscape management, and support policy and investment discussions.

The baseline profiles developed under SUSTAIN-Africa provide an important starting point for continued testing, learning and refinement. As the LRI is applied in new landscapes and institutional contexts, IUCN

will continue to refine the indicator framework and the scoring thresholds, strengthen data collection protocols, and explore how the tool can be embedded in monitoring, reporting, policy, and investment processes.

Looking ahead, IUCN will work with partners to further test the LRI across different landscapes, improve its practical guidance for users, and ensure that it remains both scientifically grounded and accessible to practitioners, communities, policy makers, the private sector, and investors. In this way, the LRI can continue to support more evidence-based, inclusive, and resilient landscape management.



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9

Annexes

Annex A: Indicator catalogue

This annex presents an updated indicator list with example sub-indicators that may be used in a landscape, along with the source of that data. This list of example sub-indicators is not exhaustive. Some tips for data collection during the selection of the sub-indicators include:

- Address Seasonality: Emphasise that dry-season sampling provides only a “partial snapshot” of land and water health. Practitioners should budget for wet-season reassessments to get a balanced view of hydrological dynamics.
- Focus on “Quick Wins”: In the implementation plan, identify low-cost actions that deliver immediate results (e.g., cleaning waste from local water sources) to build stakeholder trust and “landscape resilience literacy”.
- Appoint a Data Coordinator: Suggest that new landscapes appoint a dedicated data coordinator to track ongoing research and keep the LRI dataset current as new environmental initiatives emerge.
- Leverage Secondary Data: To reduce implementation costs, practitioners can use global datasets (e.g., Global Forest Watch for tree cover or GE-OGLOWS for river discharge) for landscape-scale indicators.



Data collection source: ¹ Primary data; ² Secondary data from the literature; ³ Secondary data from ongoing datasets.

	Kilombero	Ihemi	Sumbawanga	Barue-Vanduzi
Site-level indicator name	Site-level specific sub-indicator			
Land				
 Above-ground biodiversity	Bird species richness ² ; invertebrate spp richness ² ; mammal spp richness ²	Bird species richness ¹	Bird species richness ¹	Bird species richness ¹
 Below-ground biodiversity	Earthworm count ²	Earthworm count ¹	Earthworm count ¹	Earthworm count ¹
 Carbon stocks	Soil organic carbon ² ; above ground carbon ²	Soil organic carbon ¹ ; above ground carbon ¹	Soil organic carbon ¹ ; above ground carbon ¹	Soil organic carbon ²
 Soil fertility	Soil C:N ² ; soil pH ² ; soil total N ² ; soil available P ²	soil total N ¹ ; Soil pH ¹	soil total N ¹ ; Soil pH ¹	Soil pH ² ; soil EC ²
Water				
 Soil-water availability	Plant available water content ²	Water holding capacity ¹	Water holding capacity ¹	Soil moisture ²
 Site-level water biological activity	Macroinvertebrate assessment (miniSASS) ³	Macroinvertebrate assessment (miniSASS) ²	Macroinvertebrate assessment (TARISSfupi) ¹	Macroinvertebrate assessment (miniSASS) ¹
 Site-level water chemistry	pH ² ; EC ² ; PPM ²	pH ¹ ; EC ¹ ; ppm ¹ ; temp ¹	pH ¹ ; EC ¹ ; ppm ¹ ; temp ¹	pH ¹ ; EC ¹ ; ppm ¹ ;

	Kilombero	Ihemi	Sumbawanga	Barue-Vanduzi
Landscape-level indicator name	Landscape-level specific sub-indicator			
Water				
 Landscape water availability	River discharge ³	River discharge ³	River discharge ³	River discharge ³
 Landscape water access	Access to water for home consumption ² ;	Access to water for home consumption ² ;	Percentage of people reporting access problems ²	Access to water for home consumption ² ;
 Landscape-level water chemistry	pH ² ; EC ² ; PPM ²	pH ¹ ; EC ¹ ; ppm ¹ ; temp ¹	pH ¹ ; EC ¹ ; ppm ¹ ; temp ¹	pH ¹ ; EC ¹ ; ppm ¹
Adaptive capacity				
 Agricultural, agroforestry and tree plantation productivity	Rice ² ; sunflower ² ; soybean ² ; sugarcane smallhold ² ; sugarcane plantation: irrigated ² ; sugarcane plantation: rainfed ²	Rice ² ; sunflower ² ; soybean ²	Rice ² ; sunflower ² ; beans ² ; maize ²	Maize ² ; soybean ² ; pigeon pea ² ; green bean ² ; cabbage ² ; potato ²
 Adoption of Sustainable Land Management (SLM) practices	Percentage of SLM practices ² : soil fertility; manure; crop rotation; soil erosion; compost	Percentage of SLM practices ² : soil fertility; manure; crop rotation; soil erosion; compost	Percentage of SLM practices ² : soil fertility; manure; crop rotation; soil erosion; compost; agroforestry	Percentage of SLM practices ² : soil fertility; manure; crop rotation; soil erosion; compost
 Household poverty	PPI score ²	PPI score ²	PPI score ²	PPI score ²
 Land cover percentage	Percentage tree cover loss ³	Percentage tree cover loss ³	Percentage tree cover loss ³	Percentage tree cover loss ³
 Landscape biodiversity	Biodiversity habitat index: proportion of species (2020) ³	Biodiversity habitat index: proportion of species (2020) ³	Biodiversity habitat index: proportion of species (2020) ³	Biodiversity habitat index: proportion of species (2020) ³
 Perception of shocks and stresses	Percentage respondents impacted by flooding ² or drought ²	Percentage respondents impacted by flooding ² or drought ²	Percentage respondents impacted by flooding ² or drought ²	Percentage respondents impacted by flooding ² or drought ²
 Extreme weather: flooding + drought occurrence	Percent land area ² affected by flooding; percentage land area affected by drought	Percent land area ² affected by flooding; percentage land area affected by drought	Percent land area ² affected by flooding; percentage land area affected by drought	Percent land area ² affected by flooding; percentage land area affected by drought

	Tanzania	Mozambique
National-level indicators	National-level specific sub-indicator	
Water		
 Water quality and quantity	SDG Voluntary National Review (VNR) 6.3.2 National Climate Change Response Strategy (2021-2026)	SDG indicator 6.3: Goal 6.3 SDG indicator 6.4: Goal 6.4 The Strategy and Action Plan for the Conservation of Biological Diversity in Mozambique 2015-2035
 Access to quality water for different uses.	SDG Voluntary National Review (VNR) 1.4.1 SDG Voluntary National Review (VNR) 2.4.1 Tanzania Nationally Determined Contribution 2021	SDG indicator 6.1.1
Adaptive capacity		
 Agricultural productivity	SAGCOT Green Print 2012	Strategic Plan for the Development of the Agrarian Sector 2030
 Carbon stored in forest and agricultural land including associated emissions reductions	SAGCOT Green Print 2012 Changwon Initiative	Strategic Plan for the Development of the Agrarian Sector 2030
 Adoption of SLM practices, certification and agroecological markets	Tanzania National Ecological Organic Agriculture Strategy 2023-2030 National Climate Change Response Strategy (2021-2026)	The Strategy and Action Plan for the Conservation of Biological Diversity in Mozambique 2015-2035 Strategic Plan for the Development of the Agrarian Sector 2030
 Healthy and diverse ecosystems	SDG Voluntary National Review (VNR) 15.1.1 Bonn Challenge National Biodiversity Strategy and Action Plan 2015-2020 Strategic goal D, target 14 National Climate Change Response Strategy (2021-2026)	The Strategy and Action Plan for the Conservation of Biological Diversity in Mozambique 2015-2035 Bonn Challenge
 Resilient communities to shocks, disasters and climate change	Tanzania Draft Development Vision 2050	Strategic Plan for the Development of the Agrarian Sector 2030
 Poverty rate	SDG Voluntary National Review (VNR) 1.2:1 AGCOT/SAGCOT	Strategic Plan for the Development of the Agrarian Sector 2030; National Development Policy: SDG Voluntary National Review (VNR) Goal 1.1

Annex B: Measuring and scoring the LRI indicators

The Landscape Resilience Index (LRI) utilises a standardised scoring system to translate complex biophysical and socio-economic data into a clear, comparable profile of landscape health. Indicators are scored on a four-point scale: poor, moderate, good, or excellent.

The choice of a four-category scale is a deliberate balance between analytical usefulness and practical accessibility. This simplicity provides a “common language” that can be understood by a diverse range of stakeholders, from farmers and Water User Association members to national decision makers. This streamlined approach is particularly important at the site level, where the first point of engagement is often with farmers and extension officers who support data collection and follow-up action. By avoiding excessive scoring complexity, the LRI remains a usable, accessible tool that practitioners can easily communicate and repeat over time to monitor progress.

How thresholds are established

LRI indicators represent broad themes (e.g., “above-ground biodiversity”) divided into specific sub-indicators (e.g., “bird species richness”). Scoring thresholds for these sub-indicators are not arbitrary; they are based on a tiered evidence framework:

- **Scientific literature:** Where established ecological ranges exist. For example, thresholds for below-ground biodiversity (earthworm counts) were set using typical ranges for East and Southern Africa derived from global environmental models. Soil pH thresholds were based on the optimal range (pH 5.5–7.0) for crop growth established in agronomic literature.
- **Policy and programmatic standards:** Where national targets or specific program objectives are defined. Agricultural productivity (crop yield) thresholds, for instance, were based on SUSTAIN-Africa target yields for Tanzania and Mozambique. Water quality (miniSASS/TARISSfupi) thresholds were derived directly from the established ecological condition categories defined within those assessment protocols.

- **Expert judgement:** In cases where established literature or policy benchmarks were unavailable, professional expertise and understanding of local conditions were used to define interim thresholds. Examples include river discharge, where segment flows were compared to 30-year averages, and tree cover loss, where categories were derived from national average ranges.

It is important to view these categories as interim and adaptive starting points rather than definitive ecological states. As new scientific evidence or policy benchmarks become available, these thresholds are intended to be refined through stakeholder validation.

Indicator aggregation and final classification

To provide a coherent picture of resilience, the LRI aggregates detailed data into broader scores. Where an indicator involves multiple sub-indicator measurements, the individual scores are averaged to produce a single representative value for that indicator. The final average score is then classified into four distinct categories:

1. **Poor:** Indicates the majority of sub-indicators are failing or far below targets, signalling a high priority for intervention.
2. **Moderate:** Suggests emerging pressures or social/infrastructure vulnerabilities.
3. **Good:** Reflects generally healthy functioning conditions that support continuity and well-being.
4. **Excellent:** Represents strong resilience and ecological status.

This scoring system ensures the LRI remains a tool for adaptive management. By revealing where resilience challenges are most pronounced—for example, if a landscape has “excellent” water chemistry but “poor” water access—practitioners can prioritise interventions that align with both local needs and national policy goals.



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